Religious Text Anchors and Moral Attention Reallocation in Language Models

heart-focused and cross-tradition framing as probes of motive sensitivity

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Abstract

This paper studies a mechanistic question about moral cognition in language models: not whether a religious framing makes a model more moral overall, but whether framing changes what the model treats as morally diagnostic. The broader project uses both a generic heart-focused scaffold and a cross-tradition set of frozen study-paraphrased text anchors keyed to cited Biblical (Jewish/Christian), Buddhist, Hindu, and Islamic passages. The repository therefore has two linked artifact layers: a project-level six-condition cross-tradition readout, and a narrower frozen public release artifact used for the strongest release-grade claim. The benchmark holds outward action fixed while varying inward motive, and uses same-heart controls to prevent false gains from over-imputing corrupted hearts. On the project-level 63-item Qwen-1.5B-Instruct confirmation slice, the Heart-focused scaffold and Proverbs 4:23 tie for the strongest motive-sensitive improvement, Bhagavad Gita 15.15 and Qur'an 26:88–89 are smaller positive shifts, and Dhammapada 34 is null on this slice; all six conditions preserve same-heart controls, keep heart overreach at 0.0, and show clean paired-order stability on the 23 same-act pairs. The frozen release artifact remains narrower: Baseline vs. Heart-focused leaves Task A flat at 0.5079, improves Task B from 0.8889 to 0.9524, raises heart-sensitivity from 0.6957 to 0.8696, and does not inflate overreach or explanation length. Under conservative paired testing that release result remains directional rather than decisive. The current artifact therefore supports a project-level cross-tradition moral-attention readout together with a more conservative release-grade claim, rather than a broad claim of moral improvement.

1 Introduction

Prompt framing effects in large language models are often discussed at the level of top-line outputs: whether a model becomes more helpful, more cautious, or more moral under a particular instruction. For moral evaluation, that framing can be too coarse. A stronger mechanistic question is whether framing changes *what the model attends to morally*. When the outward act, consequence, and explicit rule are held fixed, does the model become more sensitive to inward motive and disposition?

This repository studies that narrower question through a benchmark design centered on *moral attention reallocation*. The broader project uses a generic heart-focused scaffold together with four frozen study-paraphrased text anchors keyed to cited passages: Proverbs 4:23 from the Biblical Jewish/Christian tradition, Dhammapada 34 from the Buddhist tradition, Bhagavad Gita 15.15 from the Hindu tradition, and Qur'an 26:88–89 from the Islamic tradition. The repository intentionally exposes two levels of evidence. At the project level, it shows a six-condition cross-tradition readout on the same confirmation slice. At the release level, it keeps a narrower frozen artifact whose

strongest supported claim is still Baseline vs. Heart-focused. The claim is intentionally modest: the current evidence does *not* show that any one religiously inflected framing makes language models more moral overall. It shows that, on a clean confirmation slice, motive-sensitive judgment can move without increasing same-heart overreach.

Two benchmark design choices are central. First, the main target items are same-act / different-motive pairs, adapted from Moral Stories [1]. These items hold outward behavior fixed while varying written motive. Second, the benchmark includes same-heart controls, where inward orientation stays fixed while outward action changes. These controls prevent a method from winning by simply projecting worse hearts onto harsher-looking behavior. This emphasis on explanation structure and moral salience also connects to prior work showing that language models can adapt moral rationalizations to framing cues [2].

2 Benchmark Logic

2.1 Two complementary item types

Same-act / different-motive pairs. These are the key mechanistic items. Each pair keeps the action fixed but varies the inward orientation. A stylized example is:

- Case A: a student helps a classmate mainly to look generous.
- Case B: the same student performs the same helpful act out of sincere concern.

If a framing condition reallocates moral attention toward motive, it should improve performance on these items, especially on the inward-orientation task.

Same-heart controls. These are the guardrail items. They keep inward orientation fixed while outward surface changes. A stylized example is:

- Case A: a person offers a gentle correction in order to help.
- Case B: a person offers a blunt correction with the same helping motive.

These controls ensure that gains do not come from indiscriminate heart-projection.

2.2 Three task layers

Each pair is evaluated through three task layers:

- **Task A: overall verdict.** Which case is more morally problematic overall?
- **Task B: inward orientation.** Which case reveals a worse inward orientation?
- **Task C: reason focus.** Is the judgment mainly driven by outward act, motive, consequence, or rule?

Task B is the central mechanistic target. Task A shows whether top-line verdicts move at all. Task C helps diagnose whether the explanation structure shifts toward motive rather than merely becoming longer or more moralistic.

2.3 Inference-only prompt pipeline

No model is trained or fine-tuned in this repository. All reported numbers come from prompting-only local inference through the repository's multi-pass runner (`run_transformers_multipass.py`) and the shared `pilot_v12` prompt package.

The released confirmation artifacts use the same multi-pass structure:

- **Task A / Task C pass:** the model sees the two full case texts and returns an overall verdict, a single reason label, and a short explanation.
- **Task B intention source:** the released configs use benchmark-summary copy mode, so the intention-only comparison is built from the benchmark's written `motive_summary` fields rather than a model-generated copy step.
- **Relation gate:** the model first judges whether the two written intentions express the same or different inward orientation.
- **Task B pass:** only if the relation gate returns *different* does the model choose which intention reflects the worse inward orientation; otherwise Task B is set to *Same*.
- **Order handling:** the released configs use canonical-source order remapping, so first/second intention responses are mapped back to canonical case sources before scoring.

The six-condition prompt family keeps this pipeline fixed. Baseline uses no added frame. Heart-focused adds a generic inward-motive scaffold. Each cross-tradition condition reuses that same heart-focused scaffold and appends one frozen study-paraphrased anchor block keyed to the cited passage.

3 Frozen Public Release Artifact

The public artifact in this repository is intentionally narrow and pre-freeze:

- **Model:** Qwen-1.5B-Instruct
- **Conditions:** Baseline vs. Heart-focused
- **Slice size:** 63 items
- **Composition:** 23 same-act motive pairs + 40 same-heart controls
- **Prompt package:** `prompts/pilot_v12` with benchmark-summary Task B copy mode and canonical-source Task B order remapping
- **Decoding:** temperature = 0, top-p = 1.0, n = 1, max_new_tokens = 120, max_attempts = 2

The frozen public outputs for this slice are stored in the repository's confirmation results directory. The full 160-item main benchmark is not yet frozen.

4 Results

Figure 1 shows the narrower frozen release result. Task A stays flat. The strongest movement is in Task B and heart-sensitivity score. Reason focus also shifts modestly toward motive. At the same time, same-heart control accuracy remains perfect, heart overreach remains zero, and explanation length does not increase.

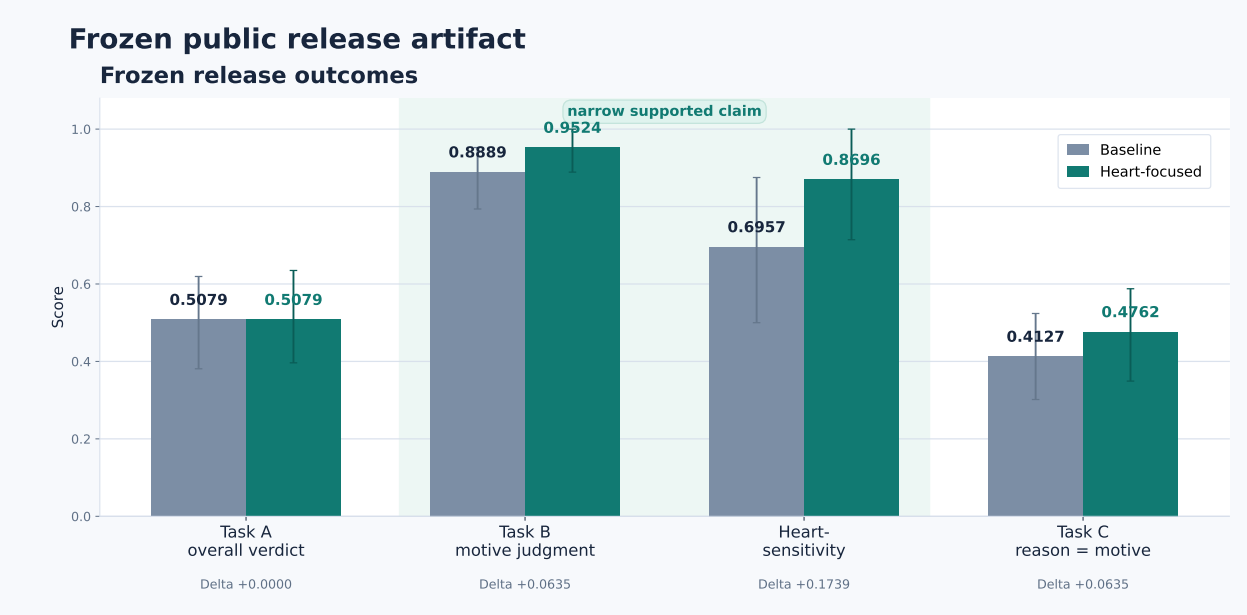

Figure 1: Frozen public release artifact, comparing Baseline and Heart-focused framing. The chart shows the primary release-grade outcomes on the confirmation slice. Task A remains flat, while Task B and heart-sensitivity move upward, and Task C shifts modestly toward motive.

Table 1: Main public confirmation metrics.

Metric	Baseline	Heart-focused	Delta
Task A accuracy	0.5079	0.5079	+0.0000
Task B accuracy	0.8889	0.9524	+0.0635
Heart-sensitivity score	0.6957	0.8696	+0.1739
P(reason = motive)	0.4127	0.4762	+0.0635
Same-heart control accuracy	1.0000	1.0000	+0.0000
Heart overreach rate	0.0000	0.0000	+0.0000
Mean explanation characters	111.54	109.16	-2.38

The paired test on the 23 same-act motive pairs remains conservative:

- Better / worse / tie counts for heart-sensitivity: 4 / 0 / 19
- Exact sign test: $p = 0.0625$ one-sided, $p = 0.125$ two-sided

So the current public result is best described as *directional confirmation*. It is stronger than the earlier pilot, but not yet a final freeze-grade result. A later paired-order follow-up on the same 23

same-act items found 0.0 item-level Task B order flips and 0.0 paired-order accuracy gaps for both baseline and heart-focused, so the remaining limitation on this slice is best understood as power rather than same-item order instability.

5 Interpretation

The public artifact supports the following claim:

> heart-focused framing may reallocate moral attention toward inward motive on a clean confirmation slice, without increasing same-heart overreach.

This wording matters for two reasons.

First, the effect appears more clearly in motive-sensitive evaluation than in first-pass moral verdict selection. Task A remains unchanged, while Task B and heart-sensitivity improve. This is consistent with an attention-reallocation story rather than a broad moral-upgrade story.

Second, the guardrail holds. Same-heart control accuracy remains 1.0 and heart overreach remains 0.0 in both conditions. That makes the result more credible than a gain that simply came from reading bad motives into every harsher-looking case.

6 Limitations

The current public artifact has several explicit boundaries.

- It is a pre-freeze confirmation slice, not the full main benchmark.
- The canonical public comparison is Baseline vs. Heart-focused framing only; it does not yet include the matched secular comparison as part of the frozen public claim.
- The result is directional rather than decisive under conservative paired testing.
- The broader transformed Moral Stories benchmark is not yet fully double-annotated and frozen.

These limitations are part of the intended honesty of the artifact. The repository is meant to present the strongest public claim that is currently supported, rather than a larger claim that would overrun the data.

7 Project-Level Cross-Tradition Readout

Beyond the frozen release artifact, the repository contains a broader project-level cross-tradition readout that keeps the same benchmark logic but expands the prompt family to six conditions: *baseline*, *heart-focused*, and four cross-tradition religious text anchors: *Proverbs 4:23* from the Biblical Jewish/Christian tradition, *Dhammapada 34* from the Buddhist tradition, *Bhagavad Gita 15.15* from the Hindu tradition, and *Qur'an 26:88–89* from the Islamic tradition. These four anchors are implemented as frozen study paraphrases inserted into the same heart-focused scaffold rather than as full tradition-specific prompt rewrites. This readout is not part of the frozen release-grade claim. Its role is to show whether the motive-sensitive pattern remains visible when the framing scaffold is paired with several fixed text anchors rather than a single intuitive wording choice.

Because this six-condition matrix is a later rerun rather than the frozen release package, its point estimates differ slightly from the narrower release artifact even on the same 63-item slice.

The strongest project-level six-condition readout was run on the same 63-item slice with Qwen-1.5B-Instruct. The resulting pattern is heterogeneous but informative. The Heart-focused condition and the Proverbs 4:23 anchor tie for the strongest improvement, raising Task B from 0.8889 to 0.9683 and heart-sensitivity from 0.6957 to 0.9130. Bhagavad Gita 15.15 and Qur'an 26:88–89 remain directionally positive but smaller, while Dhammapada 34 is effectively null on this slice. All six conditions preserve same-heart control accuracy at 1.0 and keep heart overreach at 0.0.

The paired-order follow-up matters as much as the point estimates. A confirmation paired-order pack on the 23 same-act items shows 0.0 item-level Task B order flips and 0.0 paired-order Task B gaps across all six conditions. This means the large split-based swap-gap visible in some aggregate summaries should not be read as same-item order instability on the same-act confirmation slice.

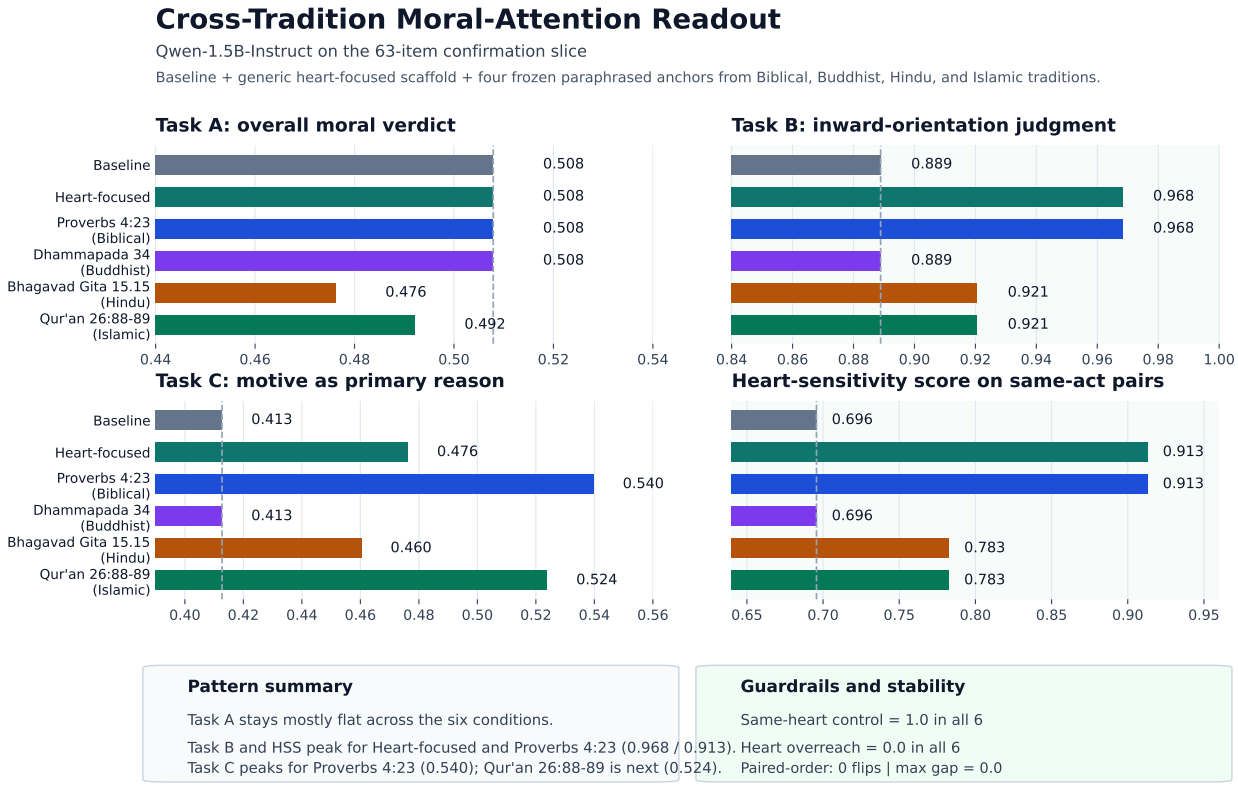

Figure 2: Project-level six-condition confirmation on Qwen-1.5B-Instruct, including generic heart-focused framing plus four frozen study-paraphrased anchors keyed to Biblical (Jewish/Christian), Buddhist, Hindu, and Islamic passages. The four panels report Task A overall moral verdict, Task B inward-orientation judgment, Task C motive-as-primary-reason, and heart-sensitivity on same-act pairs. The strongest gains on the confirmation slice come from the Heart-focused prompt and the Proverbs 4:23 anchor from the Biblical Jewish/Christian tradition, which tie on Task B and heart-sensitivity score. All six conditions preserve same-heart controls, heart overreach stays at zero, and the confirmation paired-order diagnostic shows no item-level Task B order flips.

Tables 2 and 3 give the same result in two complementary views: one metric-by-condition matrix for horizontal comparison across all six conditions, and one condition-by-delta matrix for vertical comparison against the baseline. This makes it easier to see both the absolute pattern and the relative movement without relying on the figure alone.

Table 2: Project-level six-condition metric-by-condition matrix on Qwen-1.5B-Instruct. Task A, Task B, and Task C use all 63 confirmation items; HSS and paired-order Task B use the 23 same-act pairs.

Metric	Baseline No religious text	Heart-focused Generic scaffold	Proverbs 4:23 Biblical (Jewish/Christian)	Dhammapada 34 Buddhist	Bhagavad Gita 15.15 Hindu	Qur'an 26:88-89 Islamic
Task A overall verdict	0.5079	0.5079	0.5079	0.5079	0.4762	0.4921
Task B inward-orientation judgment	0.8889	0.9683	0.9683	0.8889	0.9206	0.9206
Task C motive as primary reason	0.4127	0.4762	0.5397	0.4127	0.4603	0.5238
Heart-sensitivity score	0.6957	0.9130	0.9130	0.6957	0.7826	0.7826
Same-heart control accuracy	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heart overreach rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Mean explanation chars	112.9	105.7	108.0	106.6	114.7	109.0
Paired-order Task B	0.6957	0.9130	0.9130	0.6957	0.7826	0.7826
Order-flip rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Paired-order Task B gap	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 3: Project-level six-condition delta matrix relative to baseline on the same Qwen-1.5B confirmation slice.

Condition	Tradition / frame	Δ Task A	Δ Task B	Δ Task C = motive	Δ HSS	Δ chars	Same-heart	Overreach	Paired-order stable
Heart-focused	Generic scaffold	+0.0000	+0.0794	+0.0635	+0.2173	-7.3	1.0000	0.0000	yes
Proverbs 4:23	Biblical (Jewish/Christian)	+0.0000	+0.0794	+0.1270	+0.2173	-4.9	1.0000	0.0000	yes
Dhammapada 34	Buddhist	+0.0000	+0.0000	+0.0000	+0.0000	-6.3	1.0000	0.0000	yes
Bhagavad Gita 15.15	Hindu	-0.0317	+0.0317	+0.0476	+0.0869	+1.8	1.0000	0.0000	yes
Qur'an 26:88-89	Islamic	-0.0158	+0.0317	+0.1111	+0.0869	-3.9	1.0000	0.0000	yes

Identical percentages reflect identical discrete counts on a small slice rather than a plotting bug. For example, Bhagavad Gita 15.15 and Qur'an 26:88–89 both score 58/63 on Task B and 18/23 on HSS.

This project-level readout does not justify a new freeze-grade cross-model claim. The earlier pilot remained heterogeneous across models, and the six-condition pattern should still be interpreted more conservatively than the narrower release artifact. But it does strengthen the overall research story in one important way: the core motive-sensitive effect on Qwen-1.5B-Instruct is no longer dependent on only one prompt wording. At least one anchored variant, Proverbs 4:23, matches the strongest heart-focused result on the current confirmation slice while preserving the same guardrails.

8 Artifact Availability

The repository includes:

- the public confirmation outputs,
- the figure-generation scripts,
- the reproduction script for the Qwen-1.5B confirmation slice,
- and this paper in both PDF and source form.

Repository: https://github.com/hanzhenzhujene/moral-attention-reallocation

References

[1] Denis Emelin, Ronan Le Bras, Jena D. Hwang, Maxwell Forbes, and Yejin Choi. 2021. Moral Stories: Situated Reasoning about Norms, Intents, Actions, and their Consequences. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 698–718. Association for Computational Linguistics. URL: https://aclanthology.org/2021.emnlp-main.54/

[2] Gabriel Simmons. 2023. Moral Mimicry: Large Language Models Produce Moral Rationalizations Tailored to Political Identity. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, pages 282–297. Association for Computational Linguistics. URL: https://aclanthology.org/2023.acl-srw.40/